

# AMBA<sup>™</sup> 3 APB Protocol

v1.0

## Specification



## AMBA 3 APB Protocol Specification

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### Release Information

#### Change history

| Date              | Issue | Change                  |
|-------------------|-------|-------------------------|
| 25 September 2003 | A     | First release for v1.0  |
| 17 August 2004    | B     | Second release for v1.0 |

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# Preface

This preface introduces the *Advanced Microcontroller Bus Architecture* (AMBA) 3 *Advanced Peripheral Bus* (APB) protocol specification. It contains the following sections:

- *About this specification* on page x
- *Feedback* on page xiii.

## About this specification

This is the specification for the AMBA 3 APB protocol. All references to APB in this manual refer to AMBA 3 (not AMBA 2 or earlier versions).

## Intended audience

This specification is written to help hardware and software engineers to design systems and modules that are compatible with the APB protocol.

## Using this specification

This specification is organized into the following chapters:

### **Chapter 1** *Introduction*

Read this chapter for an overview of the APB protocol.

### **Chapter 2** *Transfers*

Read this chapter for information about the different types of APB transfer.

### **Chapter 3** *Operating States*

Read this chapter for descriptions of the APB operating states.

### **Chapter 4** *Signal Descriptions*

Read this chapter for descriptions of the APB signals.

## Conventions

This section describes the conventions that this specification uses:

- *Typographical*
- *Timing diagrams* on page xi
- *Signals* on page xii.

### **Typographical**

This specification uses the following typographical conventions:

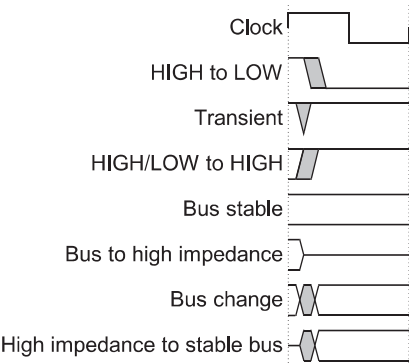
|                      |                                                                                                                                                     |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><i>italic</i></b> | Highlights important notes, introduces special terminology, denotes internal cross-references, and citations.                                       |
| <b>bold</b>          | Highlights interface elements, such as menu names. Denotes ARM processor signal names. Also used for terms in descriptive lists, where appropriate. |

|                                      |                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>monospace</code>               | Denotes text that you can enter at the keyboard, such as commands, file and program names, and source code.                                                                                                                                                                                                                                          |
| <u><code>monospace</code></u>        | Denotes a permitted abbreviation for a command or option. You can enter the underlined text instead of the full command or option name.                                                                                                                                                                                                              |
| <i><code>monospace italic</code></i> | Denotes arguments to monospace text where the argument is to be replaced by a specific value.                                                                                                                                                                                                                                                        |
| <b><code>monospace bold</code></b>   | denotes language keywords when used outside example code.                                                                                                                                                                                                                                                                                            |
| <code>&lt; and &gt;</code>           | Angle brackets enclose replaceable terms for assembler syntax where they appear in code or code fragments. They appear in normal font in running text. For example: <ul style="list-style-type: none"><li>MRC p15, 0 &lt;Rd&gt;, &lt;CRn&gt;, &lt;CRm&gt;, &lt;Opcode_2&gt;</li><li>The Opcode_2 value selects which register is accessed.</li></ul> |

Timing diagrams

The figure named *Key to timing diagram conventions* explains the components used in timing diagrams. Variations, when they occur, have clear labels. You must not assume any timing information that is not explicit in the diagrams.

Shaded bus and signal areas are undefined, so the bus or signal can assume any value within the shaded area at that time. The actual level is unimportant and does not affect normal operation.



Key to timing diagram conventions

## Signals

The signal conventions are:

|                     |                                                                                                                                                                         |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Signal level</b> | The level of an asserted signal depends on whether the signal is active-HIGH or active-LOW. Asserted means HIGH for active-HIGH signals and LOW for active-LOW signals. |
| <b>Prefix P</b>     | Denotes AMBA 3 APB signals.                                                                                                                                             |
| <b>Suffix n</b>     | Denotes AXI, AHB, and AMBA 3 APB reset signals.                                                                                                                         |

## Further reading

This section lists publications that provide additional information about the AMBA 3 protocol family.

ARM periodically provides updates and corrections to its documentation. See <http://www.arm.com> for current errata sheets, addenda, and the Frequently Asked Questions list.

This document contains information that is specific to the APB interface. See the following documents for other relevant information:

- *AMBA AXI Protocol Specification* (ARM IHI 0022).

## Feedback

ARM Limited welcomes feedback on the APB protocol and its documentation.

### Feedback on the product

If you have any comments or suggestions about this product, contact your supplier giving:

- the product name
- a concise explanation of your comments.

### Feedback on this specification

If you have any comments on this specification, send email to [errata@arm.com](mailto:errata@arm.com) giving:

- the title
- the number
- the relevant page number(s) to which your comments apply
- a concise explanation of your comments.

ARM Limited also welcomes general suggestions for additions and improvements.

*Preface*

# Chapter 1

## Introduction

This chapter provides an overview of the AMBA 3 APB. It contains the following section:

- *About the AMBA 3 APB* on page 1-2
- *Changes for AMBA 3 APB Protocol Specification v1.0* on page 1-3.

## 1.1 About the AMBA 3 APB

The APB is part of the AMBA 3 protocol family. It provides a low-cost interface that is optimized for minimal power consumption and reduced interface complexity.

The APB interfaces to any peripherals that are low-bandwidth and do not require the high performance of a pipelined bus interface. The APB has unpipelined protocol.

All signal transitions are only related to the rising edge of the clock to enable the integration of APB peripherals easily into any design flow. Every transfer takes at least two cycles.

The APB can interface with the AMBA *Advanced High-performance Bus Lite* (AHB-Lite) and AMBA *Advanced Extensible Interface* (AXI). You can use it to provide access to the programmable control registers of peripheral devices.



## 1.2 Changes for AMBA 3 APB Protocol Specification v1.0

This version includes:

- A ready signal, **PREADY**, to extend an APB transfer. See Chapter 2 *Transfers*.
- An error signal, **PSLVERR**, to indicate the failure of a transfer. See *Error response* on page 2-6.

## *Introduction*

## Chapter 2

# Transfers

This chapter describes typical AMBA 3 APB write and read transfers, and the error response. It contains the following sections:

- *Write transfers* on page 2-2
- *Read transfers* on page 2-4
- *Error response* on page 2-6.

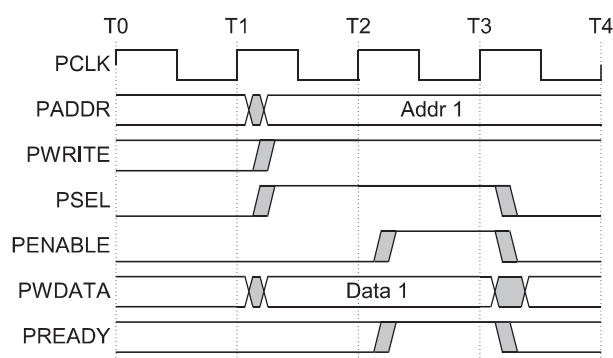
## 2.1 Write transfers

Two types of write transfer are described in this section:

- *With no wait states*
- *With wait states.*

### 2.1.1 With no wait states

Figure 2-1 shows a basic write transfer with no wait states.



**Figure 2-1 Write transfer with no wait states**

The write transfer starts with the address, write data, write signal and select signal all changing after the rising edge of the clock. The first clock cycle of the transfer is called the Setup phase. After the following clock edge the enable signal is asserted, **PENABLE**, and this indicates that the Access phase is taking place. The address, data and control signals all remain valid throughout the Access phase. The transfer completes at the end of this cycle.

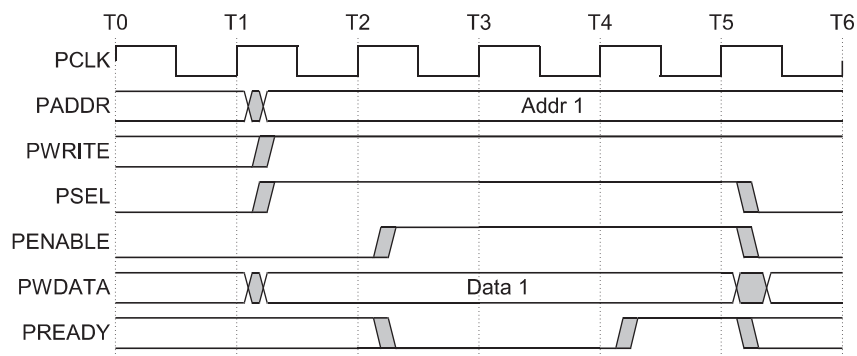
The enable signal, **PENABLE**, is deasserted at the end of the transfer. The select signal, **PSELx**, also goes LOW unless the transfer is to be followed immediately by another transfer to the same peripheral.

### 2.1.2 With wait states

Figure 2-2 on page 2-3 shows how the **PREADY** signal from the slave can extend the transfer. During an Access phase, when **PENABLE** is HIGH, the transfer can be extended by driving **PREADY** LOW. The following signals remain unchanged for the additional cycles:

- address, **PADDR**
- write signal, **PWRITE**

- select signal, **PSEL**
- enable signal, **PENABLE**
- write data, **PWDATA**.



**Figure 2-2 Write transfer with wait states**

**PREADY** can take any value when **PENABLE** is LOW. This ensures that peripherals that have a fixed two cycle access can tie **PREADY** HIGH.

**Note**

It is recommended that the address and write signals are not changed immediately after a transfer but remain stable until another access occurs. This reduces power consumption.

## 2.2 Read transfers

Two types of read transfer are described in this section:

- *With no wait states*
- *With wait states.*

### 2.2.1 With no wait states

Figure 2-3 shows a read transfer. The timing of the address, write, select, and enable signals are as described in *Write transfers* on page 2-2. The slave must provide the data before the end of the read transfer.

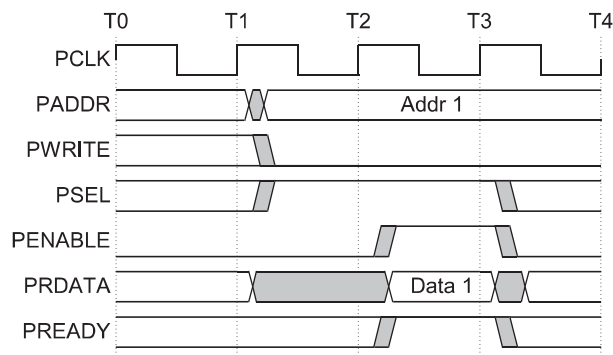


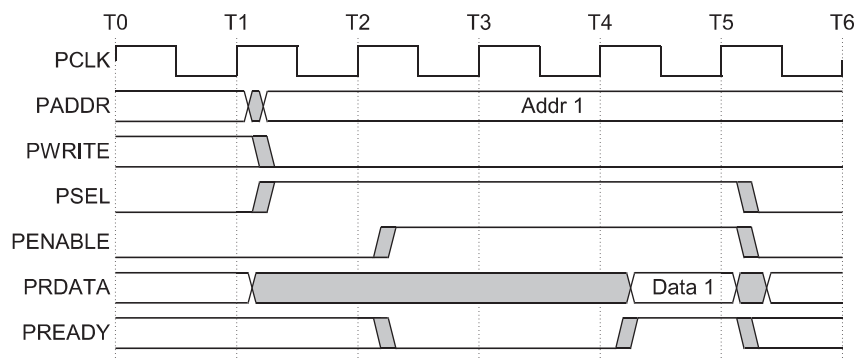
Figure 2-3 Read transfer with no wait states

### 2.2.2 With wait states

Figure 2-4 on page 2-5 shows how the **PREADY** signal can extend the transfer. The transfer is extended if **PREADY** is driven LOW during an Access phase. The protocol ensures that the following remain unchanged for the additional cycles:

- address, **PADDR**
- write signal, **PWRITE**
- select signal, **PSEL**
- enable signal, **PENABLE**.

Figure 2-4 on page 2-5 shows that two cycles are added using the **PREADY** signal. However, you can add any number of additional cycles, from zero upwards.

**Figure 2-4 Read transfer with wait states**

## 2.3 Error response

You can use **PSLVERR** to indicate an error condition on an APB transfer. Error conditions can occur on both read and write transactions.

**PSLVERR** is only considered valid during the last cycle of an APB transfer, when **PSEL**, **PENABLE**, and **PREADY** are all HIGH.

It is recommended, but not mandatory, that you drive **PSLVERR** LOW when it is not being sampled. That is, when any of **PSEL**, **PENABLE**, or **PREADY** are LOW.

Transactions that receive an error, might or might not have changed the state of the peripheral. This is peripheral-specific and either is acceptable. When a write transaction receives an error this does not mean that the register within the peripheral has not been updated. Read transactions that receive an error can return invalid data. There is no requirement for the peripheral to drive the data bus to all 0s for a read error.

APB peripherals are not required to support the **PSLVERR** pin. This is true for both existing and new APB peripheral designs. Where a peripheral does not include this pin then the appropriate input to the APB bridge is tied LOW.

### 2.3.1 Write transfer

Figure 2-5 shows an example of a failing write transfer that completes with an error.

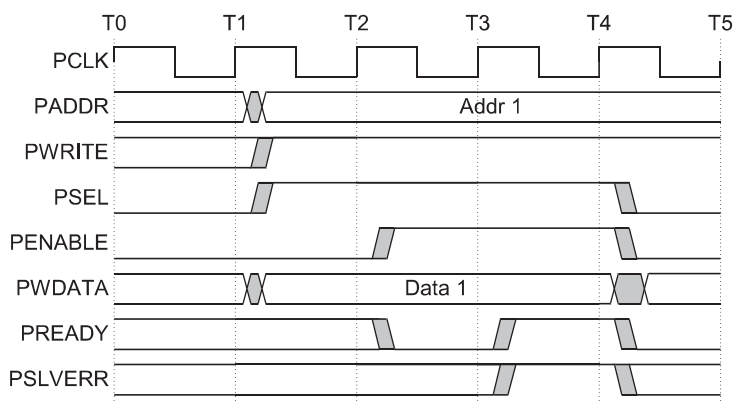


Figure 2-5 Example failing write transfer



### 2.3.2 Read transfer

A read transfer can also complete with an error response, indicating that there is no valid read data available. Figure 2-6 on page 2-7 shows a read transfer completing with an error response.

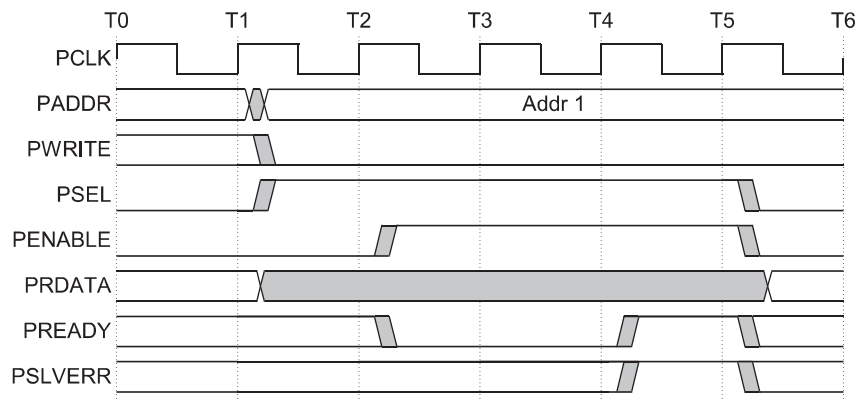


Figure 2-6 Example failing read transfer

### 2.3.3 Mapping of PSLVERR

When bridging:

**From AXI to APB** An APB error is mapped back to **RRESP/BRESP = SLVERR**. This is achieved by mapping **PSLVERR** to the AXI signals **RRESP[1]** for reads and **BRESP[1]** for writes.

**From AHB to APB** **PSLVERR** is mapped back to **HRESP = ERROR** for both reads and writes. This is achieved by mapping **PSLVERR** to the AHB signal **HRESP[0]**.

*Transfers*

## Chapter 3

# Operating States

This chapter describes the AMBA 3 APB operating states. It contains the following section:

- *Operating states* on page 3-2.

### 3.1 Operating states

Figure 3-1 shows the operational activity of the APB.

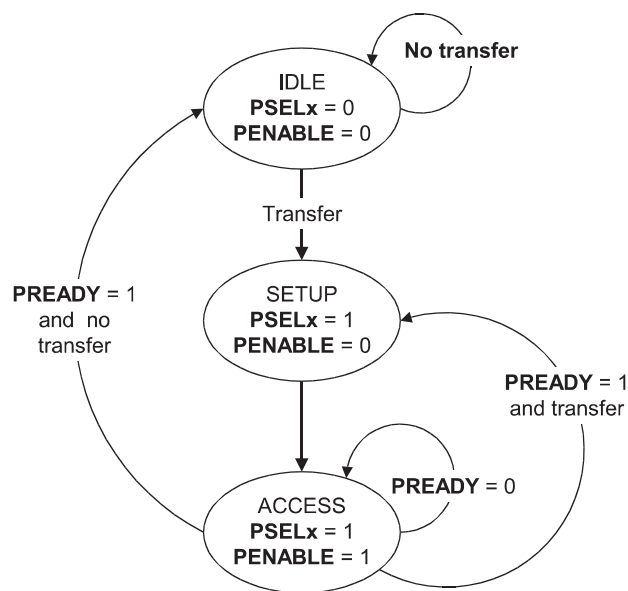


Figure 3-1 State diagram

The state machine operates through the following states:

- IDLE** This is the default state of the APB.
- SETUP** When a transfer is required the bus moves into the SETUP state, where the appropriate select signal, **PSELx**, is asserted. The bus only remains in the SETUP state for one clock cycle and always moves to the ACCESS state on the next rising edge of the clock.
- ACCESS** The enable signal, **PENABLE**, is asserted in the ACCESS state. The address, write, select, and write data signals must remain stable during the transition from the SETUP to ACCESS state.
- Exit from the ACCESS state is controlled by the **PREADY** signal from the slave:
- If **PREADY** is held LOW by the slave then the peripheral bus remains in the ACCESS state.

- If **PREADY** is driven HIGH by the slave then the **ACCESS** state is exited and the bus returns to the **IDLE** state if no more transfers are required. Alternatively, the bus moves directly to the **SETUP** state if another transfer follows.

## *Operating States*

## Chapter 4

# Signal Descriptions

This chapter describes the AMBA 3 APB signals. It contains the following section:

- *AMBA 3 APB signals* on page 4-2.

## 4.1 AMBA 3 APB signals

Table 4-1 lists the APB signal descriptions.

**Table 4-1 APB signal descriptions**

| Signal         | Source                | Description                                                                                                                                                                                                                                                                         |
|----------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PCLK</b>    | Clock source          | Clock. The rising edge of <b>PCLK</b> times all transfers on the APB.                                                                                                                                                                                                               |
| <b>PRESETn</b> | System bus equivalent | Reset. The APB reset signal is active LOW. This signal is normally connected directly to the system bus reset signal.                                                                                                                                                               |
| <b>PADDR</b>   | APB bridge            | Address. This is the APB address bus. It can be up to 32 bits wide and is driven by the peripheral bus bridge unit.                                                                                                                                                                 |
| <b>PSELx</b>   | APB bridge            | Select. The APB bridge unit generates this signal to each peripheral bus slave. It indicates that the slave device is selected and that a data transfer is required. There is a <b>PSELx</b> signal for each slave.                                                                 |
| <b>PENABLE</b> | APB bridge            | Enable. This signal indicates the second and subsequent cycles of an APB transfer.                                                                                                                                                                                                  |
| <b>PWRITE</b>  | APB bridge            | Direction. This signal indicates an APB write access when HIGH and an APB read access when LOW.                                                                                                                                                                                     |
| <b>PWDATA</b>  | APB bridge            | Write data. This bus is driven by the peripheral bus bridge unit during write cycles when <b>PWRITE</b> is HIGH. This bus can be up to 32 bits wide.                                                                                                                                |
| <b>PREADY</b>  | Slave interface       | Ready. The slave uses this signal to extend an APB transfer.                                                                                                                                                                                                                        |
| <b>PRDATA</b>  | Slave interface       | Read Data. The selected slave drives this bus during read cycles when <b>PWRITE</b> is LOW. This bus can be up to 32-bits wide.                                                                                                                                                     |
| <b>PSLVERR</b> | Slave interface       | This signal indicates a transfer failure. APB peripherals are not required to support the <b>PSLVERR</b> pin. This is true for both existing and new APB peripheral designs. Where a peripheral does not include this pin then the appropriate input to the APB bridge is tied LOW. |



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